

Lecture 09/11: 2D Systems

→ "we are free from the shackles of 1dim, where we can only march or turn around"

S6: Phase Plane (State plane/State space)

$$\begin{aligned} \dot{x} &= f(x, y) \\ \dot{y} &= g(x, y) \end{aligned} \rightarrow \dot{\vec{x}} = \vec{f}(\vec{x}) = \underline{\underline{A}} \cdot \vec{x} \quad \text{for some linear system}$$

6.1. Equilibria $\rightarrow \vec{0} = \vec{f}(\vec{x})$

2. Closed Periodic orbits(!) can exist in 2-dimensional systems $\rightarrow \vec{x}(T) = \vec{x}(t=0) = \vec{x}(t=2T)$

3. Topology: state space manifold can take a variety of topological shapes, like, e.g. a torus, or a spherical surface

→ closed / periodic orbits in state space represent trajectories that continuously vary state, but along a constrained line

→ periodic orbits are the 1D generalization of an equilibrium point of a system

S6.3 Stability of equilibria

$$\left. \begin{aligned} 0 &= f(x^*, y^*) \\ 0 &= g(x^*, y^*) \end{aligned} \right\} \text{ at an equilibrium point}$$

To consider stability of a fixed point, we assume our manifold is smooth and locally linear:

$\dot{x}^* + \dot{u} \Rightarrow$ Linearization

$$= f(x^*, y^* + u) + \partial_x f(x^*, y^*)u + \partial_y f(x^*, y^*)v$$

$$\dot{u} = f(x^*, y^*) + \partial_x f(x^*, y^*)u + \partial_y f(x^*, y^*)v + \dots O(u^2)$$

$$\dot{v} = g(x^*, y^*) + \partial_x g(x^*, y^*)u + \partial_y g(x^*, y^*)v + \dots O(v^2)$$

$= 0 @ \text{equilibrium}$

$$\Rightarrow \begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \underline{A} \cdot \begin{pmatrix} u \\ v \end{pmatrix}; \quad \underline{A} = \begin{pmatrix} \frac{\partial f}{\partial x}(x^*, y^*) & \frac{\partial f}{\partial y}(x^*, y^*) \\ \frac{\partial g}{\partial x}(x^*, y^*) & \frac{\partial g}{\partial y}(x^*, y^*) \end{pmatrix}$$

$[\underline{A}] = \frac{1}{T}$ (units of Hz, or 1 over time) Matrix of velocity gradients
 $\rightarrow$ "Stability Matrix"

$\Rightarrow$ you need a time unit to evaluate the 4 numbers

$\rightarrow$ this is similar / related to the Jacobian Matrix, via:

Jacobian for coordinate transformation
measures deformation.

$$\underline{J} = \begin{bmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{bmatrix}$$

$\rightarrow$ if you have a system changing continuously, then we have that

$$\underline{J}(t) = \exp(\underline{A}t); \text{ i.e., the Stability Matrix is}$$

the logarithm of the Jacobian Matrix, evaluated at a fixed point. This works at a fixed point since the system doesn't change; in other locations, the correspondence is not the same.

$\rightarrow$ This linearization (about a fixed point) of the system state space manifold yields the Stability Matrix; Strogatz calls this Stability Matrix the Jacobian.

S6.4) Example: Rabbits vs. Sheep

analytical population dynamics

$$\text{rabbits: } f = \dot{x} = x(3 - x - 2y)$$

$$\text{sheep: } g = \dot{y} = y(2 - y - x)$$

Malthusian growth term

logistic term

interactions (finite resources) (finite resources)

$$A = \begin{pmatrix} \frac{\partial x f}{\partial x} & \frac{\partial x f}{\partial y} \\ \frac{\partial y g}{\partial x} & \frac{\partial y g}{\partial y} \end{pmatrix}$$

→ First, find equilibrium solutions:

- obvious, trivial: $(x^*, y^*) = (0, 0)$ (1)
- only single pop:
 - only rabbits $(x^*, y^*) = (3, 0)$ (2)
 - only sheep $(x^*, y^*) = (0, 2)$ (3)
- nontrivial equilibrium: $(x^*, y^*) = (1, 1)$ (4)

→ Since the system is 2 polynomials, the max number of zeroes is $2 \times (\# \text{ of parabola zeroes}) = 4$

$$A = \begin{pmatrix} \frac{\partial x f}{\partial x} & \frac{\partial x f}{\partial y} \\ \frac{\partial y g}{\partial x} & \frac{\partial y g}{\partial y} \end{pmatrix} = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - 2y - x \end{pmatrix}$$

Evaluating Stability:

$$(1): A(0, 0) = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$$

$$(2): A(3, 0) = \begin{pmatrix} -3 & -6 \\ 0 & -1 \end{pmatrix}$$

$$(x, y) = (x^*, y^*)$$

(1) $\underline{A} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \rightarrow$ along x -axis: eigenvector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$
 eigenvalue: 3
 $\rightarrow x(t) = x_0 e^{3t}$ unstable

$\rightarrow$ along y -axis: $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$, yet e^{2t} : less so, but still unstable

$\rightarrow$ eigenvectors form orthonormal basis, of course (system is decoupled)

(2) $\underline{A}(3, 0) = \begin{pmatrix} -3 & -6 \\ 0 & -1 \end{pmatrix} \rightarrow$ Stable!
 $x \sim e^{-3t}$, $y \sim e^{-t}$

(3) $\underline{A}(0, 2) = \begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix} \rightarrow$ Stable!

The above upper/lower triangular require no manipulation

(4) $\underline{A}(1, 1) = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}$
 That changes when eq #4

$\text{Tr } \underline{A} = -2$, $\det \underline{A} = -1$

$\Rightarrow \lambda_{+,-} = -1 \pm \sqrt{2}$

$\lambda_+ = \sqrt{2} - 1 > 0$ unstable

$\lambda_- = -(1 + \sqrt{2}) < 0$ stable

$\Rightarrow$ Saddle node!

$\rightarrow$ the eigenvector corresponding to λ_+ shows the unstable direction: $\vec{v}_+$

